9.6.16 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2015

Max. Marks: 100 | **Duration: 3 hrs**
Class: SE | Semester: III
Name of the Course: Digital Logic Design and Analysis | Branch: COMP
Course Code: UCEC304

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q1(a)	Convert $(243.63)_8$ into decimal, binary and hex.	06
Q1(b)	Convert $(110101)_2$ and $(101100)_2$ into gray code.	04
Q1(c)	A receiver received the following Hamming code 0011100101101 with odd parity. Find the error in the received code and give the corrected data.	10
Q2	Design a full adder using half adder. OR Implement a 32:1 Multiplexer using 16:1 Multiplexer	10
Q3 (a)	Simplify the boolean expression using K map $f(A,B,C,D) = \prod M(1,2,3,8,9,10,11,14) \cdot d(7,15)$	05
Q3(b)	Simplify using Quine Mc Cluskey Method $f(A,B,C,D,E) = \prod M(2,7,8,9,10,12)$	10
Q3(c)	Draw the diagram of a 4 bit universal shift register and explain. OR Explain the functioning of a 4 bit Bidirectional Shift Register.	05
Q4(a)	Implement a Gray to Binary code convertor with circuit diagram. OR Design a Modulo- 5 synchronous counter using T flip flop. Un-Used states are Zeros.	10
Q4(b)	Design a 3 bit binary up/down ripple counter.	10
Q5(a)	Explain the following IC characteristics: Power dissipation, propagation delay, fan in, fan out and voltage levels	10
Q5(b)	Compare TTL and CMOS Logic families based on their IC characteristics.	10
Q6 (a)	Write the entity declaration constructs in VHDL for AND and OR gate.	04
Q6 (b)	Write a short note on behavioral modeling.	06